# Academic Paper Reviewer: A Multi-Perspective Review Skill for Computer Vision and Machine Learning Papers

Technical Report, version 2.10

Academic Paper Reviewer project
https://github.com/Shyandram/academic-paper-reviewer.skill

November 2026

**Abstract**

We describe *Academic Paper Reviewer*, a Markdown skill for Claude and Codex agents that simulates an editorial process for a manuscript. The process has a field-aware panel of five independent reviewers, a literature check that cannot find the paper under review, a report check, and an Editor-in-Chief (EIC) who weights each review by its quality and makes the decision. Scores are aggregated by a deterministic, standard-library Python script and translated into the forms of 11 venues.

We evaluate the skill in two ways.

- **Accepted papers with planted flaws.** On 30 recently accepted CVPR/ECCV/ICLR papers, 96% of cited locations, 99% of quoted phrases and all but two of 842 quoted numbers are found in the paper, and the reviewers mention all six planted fatal flaws.
- **ICLR 2026 calibration.** This follows the paperreview.ai protocol. On a 20-paper stratified pilot with Sonnet at high effort, the reviewers' mean rating reaches Spearman 0.48 [0.09, 0.74] with the mean human rating and an acceptance AUC of 0.67. On the ICLR scale the normalized mean absolute error is 0.079, with essentially no bias (NME +0.004).

A smaller model (Haiku) shows no reliable signal. We also report a negative result: NMAE alone does not measure skill on this pool, because a constant predictor scores 0.085. All numbers here come from pilots. The full balanced evaluation is pending.

## 1 Introduction

Large language models can now write fluent paper reviews. Fluency, however, is not the goal of a review. Authors need the decisive weaknesses found, grounded in the manuscript, and ranked by how much they matter at a target venue. Three failure modes are common:

- **Ungrounded criticism:** weaknesses that the paper already answers, or locations and numbers that do not exist.
- **Compressed scores:** every paper rated "borderline".
- **Leakage:** the reviewer finds the paper's own camera-ready version, reviews or acceptance status, through retrieval or memory.

Academic Paper Reviewer addresses these with structure rather than with a single prompt.

- Several reviewers with distinct roles write independently.
- A Devil's Advocate attacks the paper.
- A report check removes ungrounded points.
- An EIC acts as area chair.

The skill is tuned for computer vision, deep learning and ML papers, with task-specific checklists for detection, segmentation, 3D/NeRF/3DGS, restoration, generation, VLMs, video, medical imaging and efficient models, but it can be used in any field. Its output is a *simulation* to help authors improve a paper, not a prediction of a venue's decision.

**Contributions.**

1. A review workflow with independent role-based reviewers, a quality-weighted EIC, and deterministic score aggregation into venue-specific forms (Section 2).
2. Safeguards against prompt injection and self-leakage in literature search (Section 3).
3. A reproducible calibration harness following the paperreview.ai protocol, extended with CSPaper-style error metrics and a no-split pilot mode (Section 5).
4. Pilot results, including what the skill did badly and the fixes that followed.

# 2 System design

## 2.1 Workflow

A full review runs the following phases. The phases are Markdown instructions in `SKILL.md` and `agents/*.md`; scripts are optional helpers.

**Phase 0: field analysis.** A field analyst identifies the paper's sub-field, technique, paper type (empirical, theoretical, dataset, system, benchmark, analysis, position) and target venue, and configures five reviewers with specific expertise. An example is "token pruning + latency benchmarking" rather than "deep learning". `integrity_scan.py` flags text addressed to reviewers or AI systems, as well as hidden characters.

**Phase 1: independent reviews.** Five reports are written independently: Methodology (R1), Domain (R2), Perspective (R3), Generalist (R4) and a Devil's Advocate (DA). In Claude Code they can run as parallel subagents. The Methodology reviewer runs an error audit in nine categories (result inconsistencies, equation and proof errors, assumptions, protocol flaws, claim distortion, inconsistent definitions, citation errors, omissions, figure/table integrity). It works from a shared factual *paper memory*, and every error needs a location, a verbatim quote and a concrete check.

**Phase 1.5: literature check.** This runs only with network access and after the reports are drafted. Searches are driven by contribution claims and benchmarks: a novelty verdict per claim, a baseline scout, a coverage pass, a field snapshot and a similarity ranking. Section 3 describes how the paper itself is excluded.

**Phase 1.6: report check.** Before the EIC reads anything, vague weaknesses, points the paper already answers, and claims the paper contradicts are removed. Cosmetic issues are grouped as Minor.

**Phase 2: EIC meta-review.** The EIC reads every report and checks the decisive points against the paper. It assigns each reviewer a quality weight in $[0.5, 1.5]$, states its own pool position, and decides. Each DA-CRITICAL issue gets an area-chair test ("would I reject on this alone?").

Other modes reuse the same components: quick assessment, methodology focus, guided (Socratic) review, rebuttal drafting and re-review of a revision.

## 2.2 Scores

Every non-DA reviewer gives NeurIPS-style 1–4 ratings, an Overall 1–6 recommendation (6 Strong Accept … 1 Strong Reject), a confidence 1–5, and seven 0–100 dimension scores: technical rigor, clarity, novelty, significance, experimental validation, reproducibility and ethics. The DA gives the 0–100 set only. Dimensions are combined with paper-type weights (Table 1).

Two score lists are kept. The **official** list holds the reviewers' raw ratings and their plain means, which is what authors see; it is never modified. The **EIC assessment** weights reviewer $i$ by

$$w_i = c(\text{conf}_i) \cdot q_i, \qquad c(5..1) = (1.00, 0.85, 0.70, 0.55, 0.40), \quad q_i \in [0.5, 1.5], \tag{1}$$

where $q_i$ is the EIC's quality weight. The decision starts from the EIC-weighted mean of the Overall ratings, rounded half up. The weighted 0–100 composite is a cross-check: it maps to

Table 1: Paper-type weights of the seven 0–100 dimensions (each row sums to 1).

| Type | Rigor | Clarity | Novelty | Signif. | Exp. val. | Repro. | Ethics |
|---|---|---|---|---|---|---|---|
| empirical | .20 | .10 | .15 | .15 | .20 | .10 | .10 |
| theoretical | .30 | .10 | .20 | .15 | .10 | .05 | .10 |
| dataset | .15 | .10 | .10 | .20 | .15 | .15 | .15 |
| system | .20 | .10 | .10 | .20 | .20 | .10 | .10 |
| benchmark | .15 | .15 | .10 | .20 | .20 | .15 | .05 |
| analysis | .25 | .10 | .20 | .20 | .15 | .05 | .05 |
| position | .15 | .20 | .25 | .25 | .05 | .00 | .10 |

Overall at thresholds 85/70/60/50/35. Two further rules apply:

- **DA cap.** Upheld DA-CRITICAL issues cap the decision at 4 if they can be resolved by clarification and at 3 otherwise.
- **Consistency warnings** flag disagreements that the EIC must explain.

**Calibration to the submission pool (2.10).** Each reviewer and the EIC first state a *pool position* $P$: the share of the venue's submissions the paper is better than, using real base rates (about 25% acceptance at major ML/CV conferences; ICLR 2026 about 27% accepted, 45% rejected, 24% withdrawn, 5% desk rejected). They then score within the matching band:

| $P$ | $\geq 95$ | 85–94 | 70–84 | 55–69 | 30–54 | $< 30$ |
|---|---|---|---|---|---|---|
| composite | 85+ | 73–84 | 62–72 | 55–61 | 42–54 | $< 42$ |

A composite of 62 is the accept-side bar. Uncertainty is never a reason to score a paper as borderline. The aggregation script warns when a composite lies more than 5 points outside its pool band, and when all reviewers sit in the borderline band.

### 2.3 Venue forms

Internal scores are translated into 11 venue forms: CVPR, ICCV, ECCV, NeurIPS, ICLR, ICML, AAAI, ACM MM, TMLR, IEEE TPAMI and IEEE TIP. Each translation applies the venue's emphasis weights, bar offset and scale. Each scale is labeled *official*, *confirmed*, *secondary*, *assumed* or *unpublished*. For ICLR 2026 the Overall maps to the rating set $\{0, 2, 4, 6, 8, 10\}$ as $r = 2(x-1)$. The legacy (2025) set is $\{1, 3, 5, 6, 8, 10\}$.

## 3 Safeguards

- **Prompt injection.** The manuscript is treated as data, never as instructions. Injected text is reported to the editor as an integrity concern and never raises a score.
- **Self-leak in literature search.** A fingerprint filter refuses queries that could find the reviewed paper. It drops the paper's own arXiv versions, code pages, reviews and acceptance lists unread, and labels every other result as prior (peer-reviewed or 6+ months old), recent preprint, concurrent or newer. Only prior work can be a required comparison. Each run logs the mode (`on`, `strict` or `off`). In `strict` mode a cutoff date applies; in evaluation this is the submission deadline.
- **Recognition.** Reviewers report when they recognize a paper from prior knowledge, and such papers are reported separately.
- **Integrity.** Reviewer personas are fictional, no citations may be fabricated, and every point is tied to a location in the manuscript.

Table 2: Grounding on 30 accepted papers: share of cited items not found in the paper. Numbers the reviewers recomputed themselves (525) are excluded and were spot-checked by hand.

| Check | Checked | Not found | Rate |
|---|---|---|---|
| Cited locations (tables, figures, sections, equations) | 1101 | 48 | 4% |
| Quoted phrases | 778 | 9 | 1% |
| Quoted numbers | 842 | 2 | 0% |

# 4 Evaluation I: accepted papers and planted flaws

An earlier release of the skill (before 2.10) reviewed 30 recently accepted papers: 10 each from CVPR, ECCV and ICLR, spread over 10 topics. All runs completed with every required section; the median review length was 10,214 words. Table 2 reports grounding.

**Planted flaws.** In six accepted papers we inserted one sentence admitting a fatal evaluation flaw: hyperparameters tuned on the test set, or the test split merged into training.

- The reviews mentioned the flaw in 6/6 papers.
- The Devil's Advocate raised it as DA-CRITICAL in 2/6, and the EIC upheld both.
- The final decision was $\leq 3$ in 5/6.

The miss was an ICLR paper rated 4.

**Decisions.** 16/30 accepted papers landed on the accept side (Overall $\geq 4$); 14 received 3. This compression motivated the pool-position calibration of version 2.10. The DA raised CRITICAL issues on 11/30 papers, and the EIC upheld none on real papers.

# 5 Evaluation II: ICLR 2026 calibration

## 5.1 Protocol

We follow the paperreview.ai protocol.

- Sample ICLR 2026 submissions with human reviews, including withdrawn ones.
- Fit an ordinary least-squares map from the seven dimension scores to the human rating on a train split.
- Report on the test split: Spearman with one human rating, and the AUC of the score for acceptance. Confidence intervals come from bootstrap resampling; results are given for all scored papers and without withdrawn papers.

The harness (`evals/iclr_calibration.py`, standard library only) runs in GitHub Actions through the API. Reviewers never see human scores or decisions.

**Error metrics.** Following the CSPaper benchmark, for predictions $\hat{s}_j$ and human ratings $s_j$ on a scale $[S_{\min}, S_{\max}]$ (ICLR 2026: $[0, 10]$),

$$\mathrm{NMAE} = \frac{1}{n}\sum_j \frac{|s_j - \hat{s}_j|}{S_{\max} - S_{\min}}, \qquad \mathrm{NME} = \frac{1}{n}\sum_j \frac{\hat{s}_j - s_j}{S_{\max} - S_{\min}}. \tag{2}$$

We report each with its per-paper standard deviation and a bootstrap 95% CI. Pilots are too small to fit the regression, so the `pilot` command scores the skill's own outputs as they are, without any fitting to human scores:

- the *panel rating*: the reviewers' mean Overall, mapped to the ICLR scale by $2(x - 1)$;
- the *final rating*: the integer decision on the ICLR scale.

Table 3: Ranking on the 20-paper pilot: Spearman with the mean human rating [bootstrap 95% CI] / acceptance AUC. Human references are on the same papers unless noted.

| Run | Panel rating (1–6) | EIC pool position | Mean of 6 dimensions |
|---|---|---|---|
| Haiku, default | +0.20 [−0.21, 0.58] / 0.56 | −0.06 [−0.55, 0.42] / 0.42 | −0.01 [−0.52, 0.49] / 0.49 |
| Sonnet, high | +**0.48** [0.09, 0.74] / **0.67** | +0.30 [−0.14, 0.69] / 0.55 | +0.22 [−0.27, 0.64] / 0.57 |
| Without the 3 papers Sonnet recognized ($n = 17$): | | | |
| Haiku, default | +0.07 / 0.50 | −0.25 / 0.33 | −0.10 / 0.44 |
| Sonnet, high | +0.46 [0.09, 0.74] / 0.69 | +0.22 / 0.53 | +0.10 / 0.54 |
| References: one human vs another +0.12, one-human AUC 0.58 (same papers); our ICLR 2026 test split 0.25 / 0.75; paperreview.ai (ICLR 2025) AI 0.42 / 0.75, human 0.41 / 0.84. | | | |

Table 4: Error on the ICLR 2026 scale (0–10) against the mean human rating, 20 papers. NMAE ± std [95% CI]; NME ± std.

| Predictor | NMAE ↓ | NME | Spearman | AUC |
|---|---|---|---|---|
| Haiku, panel rating | $0.093 \pm 0.057$ [0.068, 0.118] | $+0.030 \pm 0.105$ | +0.21 | 0.56 |
| Haiku, final rating | $0.085 \pm 0.063$ [0.058, 0.115] | $-0.020 \pm 0.104$ | n/a (constant) | 0.50 |
| Sonnet, panel rating | $\mathbf{0.079} \pm 0.061$ [0.054, 0.106] | $\mathbf{+0.004} \pm 0.100$ | **+0.48** | **0.67** |
| Sonnet, final rating | $0.090 \pm 0.064$ [0.061, 0.119] | $+0.010 \pm 0.110$ | +0.26 | 0.53 |
| Constant at the median human mean | $0.085 \pm 0.060$ | – | – | – |
| One human vs another human | $0.180 \pm 0.125$ | – | +0.12 | – |
| One human vs mean human | $0.102 \pm 0.065$ | – | – | – |

## 5.2 Data and a decision leak

The sample is ICLR 2026 with seed 2025: 300 papers drawn, 295 with human scores. Our first pilot used the PDFs from paperreview.ai's dump. These turned out to leak the decision: every accepted paper was a camera-ready version (author names, acknowledgments, no line numbers), while all other papers were anonymous submissions. Overall, 0/29 accepted papers and 56/56 other papers showed "Anonymous authors". We therefore use only arXiv versions posted before the 2026-01-26 decision date, normalized identically. The header before the abstract is replaced by the title, and line numbers, arXiv stamps, emails, affiliation footnotes, code links, acknowledgments and acceptance marks are removed. The literature check runs in `strict` mode with the submission deadline as cutoff.

The pilot set has 20 papers stratified by decision and human score: 5 accepted, 10 rejected, 4 withdrawn and 1 desk rejected.

## 5.3 Pilots

| Pilot | Skill | Model / effort | Outcome |
|---|---|---|---|
| 1 | 2.9 | Haiku / default | decision leak in the text; scores compressed; Spearman −0.12, AUC 0.47 |
| 2 | 2.10 | Haiku / default | normalized text; Spearman −0.01, AUC 0.49 (mean of 6 dimensions) |
| 3 | 2.10 | Sonnet / high | normalized text; Tables 3–4 |

In pilot 1 nearly every paper received Overall 3, including papers with human means of 2.0 and 6.0. The anchors described papers only relative to *accepted* papers, and the anti-harshness rule could only raise scores. Version 2.10 added pool positions, base rates, two-sided checks and EIC rulings on borderline panels.

### 5.4 Findings

- **Model capability matters.** Haiku's scores did not separate papers. Its integer final rating was the same for all 20 papers, so Spearman is undefined. Sonnet at high effort gave the first signal whose CI excludes zero, at about the level of human–human agreement.
- **NMAE rewards the middle.** An oracle constant at the median human mean scores 0.085. That beats one human against another (0.180), and Haiku's constant output scores the same. NMAE must therefore be read together with a ranking metric. Only Sonnet's panel rating beats the constant *and* ranks papers.
- **No bias after calibration.** NME is close to zero for both models, so the 2.10 bands no longer shift scores up or down on this pool.
- **The integer decision loses signal.** Rounding the panel mean to an integer decision lowers Spearman from 0.48 to 0.26 for Sonnet. For ranking, the continuous panel rating should be the primary score.
- **Recognition.** The three papers Sonnet recognized all scored high (EIC pool position 65–70) regardless of outcome. Removing them changes the Sonnet results little.
- **Cost.** About $0.86 per paper for Haiku and $1.97 per paper for Sonnet at high effort.

For reference only: CSPaper reports NMAE 0.128, NME −0.015 and SRC 0.658 for Claude Sonnet 5.5 on ICLR main. That benchmark's sample size and paper overlap are unknown, so the numbers are not directly comparable with ours.

## 6 Related benchmarks

**paperreview.ai.** Score agreement and acceptance AUC on ICLR; our protocol follows it.

**CSPaper.** A score-prediction leaderboard with NMAE, NME, SRC and average word count; we adopt its error metrics.

**CoCoReviewBench.** (Deng et al., arXiv 2605.07905, CC BY 4.0.) This benchmark scores review *content* rather than scores. It covers 3,900 ICLR 2017–2025 and NeurIPS 2021–2024 papers with human references filtered for correctness through reviewer–reviewer and reviewer–author conflicts, using the meta-review as context. An LLM judge scores correctness, thoroughness, grounding, verifiability and clarity per category, and the benchmark also reports category completeness. Its findings are that lexical metrics reward overlap rather than correctness, that reasoning models match or beat humans on grounding and verifiability, and that AI reviews are broad but shallow. Integrating it needs a GPU-served split/classify model and a judge API; its paper text must also be blinded, and every year predates current training cutoffs.

**IntelLabs AI-Peer-Review-Detection-Benchmark.** About 789k paired human and AI reviews for *detecting* AI-written reviews. It has a different goal and a restrictive license (internal R&D only), so we do not use it.

## 7 Limitations and future work

- **Small samples.** With 20 papers the intervals are wide, so these are pilot numbers. The planned balanced evaluation has 86 papers (45 train, 41 test) and fits the seven-dimension regression.
- **One venue and one year.** Calibration covers ICLR 2026 only. Other venues use assumed or secondary scales.
- **Memorization.** Strong models recognize some papers. We report recognized papers separately but cannot detect unreported recognition.
- **Scores are not the review.** Ranking agreement says nothing about whether the stated weaknesses are correct. A content-level evaluation (for example CoCoReviewBench) is the natural next step.

- **Human labels are noisy.** On our pilot papers, one human agrees with another at only Spearman 0.12. Mean human ratings are a weak target, and acceptance is a coarse one.

## Reproducibility

The skill, the scoring script, the evaluation harness, per-paper scores and paper IDs are in the repository. Manuscripts and review text are not committed. Key commands:

```
python3 -m unittest discover -s tests -v
python3 evals/iclr_calibration.py features RESULTS_DIR --out features.jsonl
python3 evals/iclr_calibration.py pilot sample.jsonl features.jsonl
python3 evals/iclr_calibration.py evaluate sample.jsonl features.jsonl \
        --splits splits.json
```

Paid runs use the workflow `.github/workflows/iclr-calibration.yml`, configured by `evals/iclr_run.json`.

## A Per-paper pilot scores

| | | Human | Haiku | | Sonnet high | | |
|---|---|---|---|---|---|---|---|
| Paper | Decision | mean | panel | pool | panel | pool | Recognized |
| AIKURJjxXT | Withdrawn | 2.50 | 3.2 | 40 | 2.8 | 40 | |
| KtocialdxS | Reject | 2.67 | 3.4 | 48 | 3.0 | 45 | |
| 8xcmfwomnI | Reject | 3.00 | 3.0 | 40 | 3.0 | 42 | |
| g4HpTtgsn2 | Withdrawn | 3.00 | 3.2 | 46 | 2.6 | 38 | |
| E5mL07Fbq8 | Reject | 3.33 | 3.0 | 38 | 2.6 | 42 | |
| Z6z4AnXGhw | Reject | 3.50 | 3.4 | 50 | 3.2 | 50 | |
| gUw8n109ki | Withdrawn | 3.50 | 3.2 | 50 | 2.6 | 40 | |
| 0xObmDmsKY | Reject | 4.00 | 3.2 | 46 | 3.0 | 45 | |
| BxeJL0rKDF | Withdrawn | 4.00 | 3.2 | 45 | 3.2 | 45 | |
| Jw7khYzYzl | Reject | 4.00 | 3.6 | 45 | 4.0 | 70 | ✓ |
| 9jcB40wjk3 | Accept | 4.40 | 2.8 | 40 | 2.8 | 38 | |
| be4ey66fhk | Reject | 4.40 | 3.2 | 45 | 3.4 | 52 | |
| 9R2iUHhVfr | Reject | 4.50 | 3.4 | 50 | 3.6 | 65 | ✓ |
| PJdMrK79Mo | Accept | 4.50 | 3.6 | 48 | 3.4 | 50 | |
| qfW4IrUuig | Reject | 4.50 | 3.0 | 40 | 2.8 | 40 | |
| LvkbeOCgmZ | Reject | 5.00 | 3.4 | 40 | 3.2 | 45 | |
| tlYSbw5GXY | Accept | 5.20 | 3.2 | 38 | 3.4 | 55 | |
| euMVC1DO4k | Accept | 6.00 | 3.6 | 50 | 3.8 | 68 | ✓ |
| kwro5432AI | Desk rej. | 6.00 | 3.2 | 45 | 3.0 | 42 | |
| rubeJmT1XM | Accept | 6.00 | 3.2 | 40 | 3.0 | 40 | |

Human mean is on the ICLR 2026 scale (0–10). "Panel" is the reviewers' mean Overall (1–6) and "pool" is the EIC pool position (0–100).